

Algebra 1 — Homework 1

Source: Tutorial sheet 1 (Tutorials/Tutorial 1.pdf) Opened 11 Aug 2026 Due Thu 13 Aug 2026, 11:59 PM

Submit: problems 0 (optional), 4, 6. TeX encouraged, not mandatory. Your submission: Homework/Submissions/HW 1 - Lilian.pdf

0 (optional). Artin, Chapter 1, problems 2.2 and 2.3. “This is the baseline.”

4. Artin, Chapter 1, problem 2.10.

6. Prove that for a subset S of $\mathbb{R}^3$, the following are equivalent.

(a) There are scalars a, b, c , not all zero, such that $S = \{(x, y, z) \mid ax + by + cz = 0\}$.

(b) There are two non-proportional vectors v, w such that $S = \{pv + qw \mid p, q \in \mathbb{R}\}$.

What is a lower-dimensional version of this? Maybe it helps to formulate and prove that first. Can you formulate a higher-dimensional version?